

Norman (1979) radiative transfer model modified to include stems

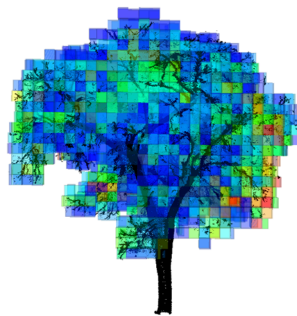
Model Documentation

Martin Béland

Digital Forest Lab

Laval University, Quebec, Canada

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DIGITAL FOREST LAB

digitalforestlab.ulaval.ca

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1. Model usage conditions

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PI contact: martin.beland@scg.ulaval.ca

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Paper to cite: Béland, M., Bonan, G. B., Kobayashi, H., and Baldocchi, D.: Consideration of radiation absorption by stems in forests for microclimate modeling, Geoscientific Model Development, in review.

2. Model usage

This code is written in Matlab. The model, including all equations, is described in Béland, M., Bonan, G. B., Kobayashi, H., and Baldocchi, D.: Consideration of radiation absorption by stems in forests for microclimate modeling, Geoscientific Model Development, in review.

The model code requires input files describing the vertical profiles for leaf and wood area index, and the leaf clumping factor. Example input files are provided for 4 plots. The model code uses a tridiagonal solver adapted from: Bonan, G.: Climate change and terrestrial ecosystem modeling, Cambridge University Press 2019.